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Polyaniline-Intercalated Flexible Two-Dimensional Metal-Organic Framework as a High-Efficiency Zn²⁺ Transport Cathode for Aqueous Zn-ion battery --Manuscript Draft--

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Thank you for your useful comments and suggestions on the language and structure of our manuscript.

We have modified the manuscript accordingly, and detailed corrections are listed below point by point:

1. Please provide the corresponding author's e-mail address in the first page of the manuscript. We have placed the corresponding author's email address on the first page of the manuscript. Thank you.

2. Abstract should be written in present tense

We have adjusted the tense in the abstract. In addition, we have carefully revised the entire manuscript and made every effort to avoid any grammatical or tense errors.

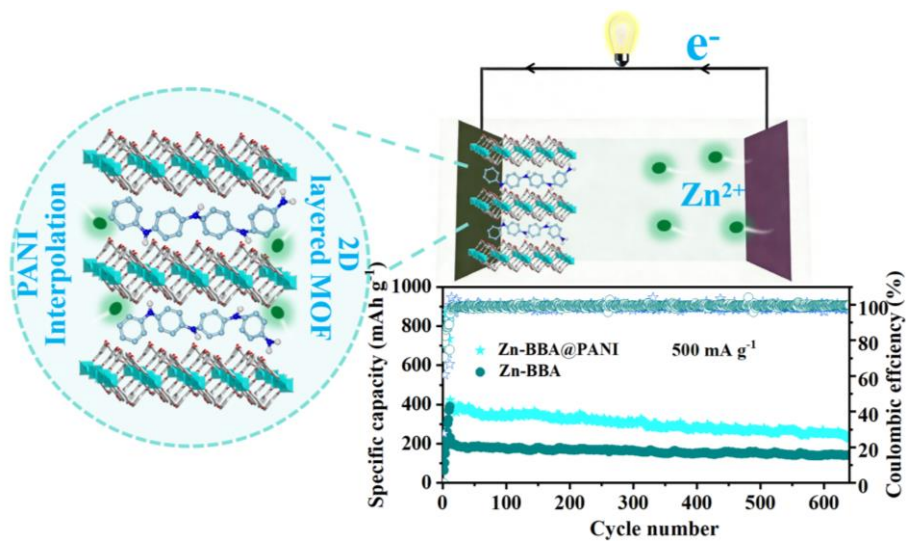
The manuscript has been resubmitted to your journal. We look forward to your positive response.

Sincerely,

<Shuangyan Wu>

Graphical abstract

PANI offers high conductivity and abundant active sites. PANI intercalation into Zn-BBA expands the interlayer spacing of the material, thereby inducing its amorphization and generating abundant active sites. Expanded interlayer spacing facilitates Zn^{2+} transport, offering new design insights for MOF-based cathodes.



Polyaniline-Intercalated Flexible Two-Dimensional Metal-Organic Framework as a High-Efficiency Zn^{2+} Transport Cathode for Aqueous Zn-ion battery

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Keywords: Aqueous zinc ion battery, Polyaniline intercalation, Composite material design, Two-dimensional coordination polymers, Interlayer spacing engineering

Abstract

Layered two-dimensional (2D) metal-organic frameworks (MOFs) are regarded as promising cathode materials for aqueous zinc-ion batteries due to their open 2D ion-transport channels, yet their intrinsic poor conductivity severely limits practical application. In this study, polyaniline (PANI) with a unique π -conjugated structure is intercalated into flexible Zn-BBA MOFs (H₃BBA = 3,5-bis-(3-carboxyphenoxy) benzoic acid), producing a Zn-BBA@PANI composite with enlarged interlayer spacing. The synergistic effect between PANI and layered MOFs effectively promotes ion migration kinetics, enhances electrical conductivity, and facilitates rapid charge transfer. The cathode exhibits outstanding electrochemical performance, after 640 cycles at a current density of 500 mA g⁻¹, it delivers a specific capacity of 219.4 mAh g⁻¹. Furthermore, it shows a reversible capacity of 189.3 mAh g⁻¹ at a high current density of 1 A g⁻¹, with excellent rate capability. This study provides an effective strategy for enhancing the ion transport and electrochemical performance of 2D layered MOF cathode materials through interlayer spacing regulation and conductive polymer integration.

1.Introduction

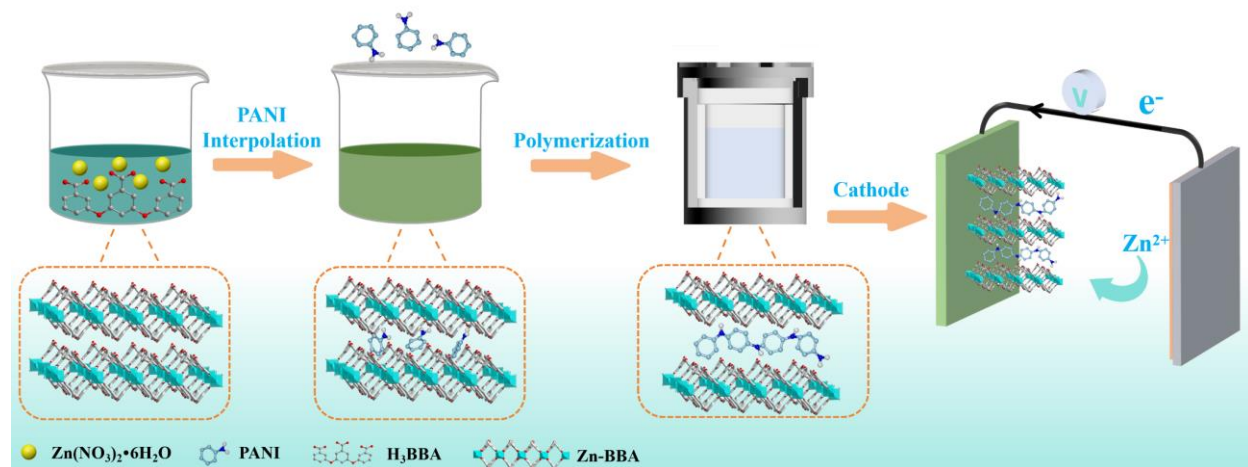
With the global transition toward renewable energy sources and the surging demand for electric vehicles, there is an urgent need to develop large-scale energy storage systems (ESSs) featuring high safety, low cost, and long lifespan.[1–8] Layered cathode materials for aqueous zinc-ion batteries have attracted significant attention in recent years due to their open 2D ion transport channels, which facilitate the efficient and reversible insertion/extraction of Zn²⁺. [9] Yet, their inherent electronic conductivity is generally low, which limits charge transfer efficiency and thereby compromises rate capability.

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4 To enhance the conductivity of such materials, composites with conductive polymers have
5 emerged as an attractive research paradigm, creating hybrid materials that combine high specific
6 surface area with excellent conductivity.[10,11] Currently, conventional strategies for modifying
7 2D layered materials with conductive polymers primarily include surface coating, physical
8 blending, and intercalation.[12,13] Compared with the former two methods, intercalation achieves
9 polymerization by directly inserting polymer monomers into the interlayer galleries of layered
10 structures, endowing the layered material with efficient 3D electron transport capability as a
11 whole.[14–17] This intrinsic enhancement of conductivity is far more thorough and stable
12 compared to external modifications, such as surface coating or physical blending, which tend to
13 result in uneven coating layers or weak interfacial adhesion. Moreover, polymer polymerization
14 within the interlayers generates effective support, expanding the interlayer spacing to create
15 spacious pathways for rapid Zn^{2+} deintercalation.[18–21] Additionally, post-intercalation,
16 interlayer interactions are partially weakened, inducing partial amorphization of the material.[22–
17 24] This process may endow the hybrid material with additional active sites.

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Currently, inorganic intercalation materials represented by conductive polymer-intercalated
layered vanadium-based and manganese-based oxides have been extensively investigated.[25–27]
Although their rigid crystal structures can provide certain ion migration pathways, the repeated
insertion/extraction of Zn^{2+} often triggers structural pulverization and capacity fading due to severe
volume variations, which restricts their long-term cycling performance.[28] In contrast to
traditional inorganic oxides, 2D layered MOFs constructed with flexible ligands exhibit
remarkable structural flexibility and toughness.[29–31] Their flexible frameworks can effectively
buffer the substantial internal stress induced by ion intercalation through reversible bond-angle
rotation or interlayer spacing adjustment. This characteristic fundamentally circumvents the

common issue of structural pulverization observed in rigid inorganic layered oxides, ensuring the structural integrity of the electrode over long-term cycling and significantly enhancing the overall specific capacity. On this basis, the introduction of small-molecule conductive polymers such as polyaniline (PANI) can further enhance the rate capability of the system. Owing to its small molecular size, PANI can readily intercalate into the interlayers of flexible 2D layered MOFs.[32–35] Through its π -conjugated structure, it forms strong interactions with the flexible MOFs, which include π - π stacking and hydrogen bonding, thereby establishing a stable heterointerface and enabling efficient electron transport.

Herein, this study employs PANI molecules with a unique π -conjugated structure and small molecular size, incorporating them into a 2D layered MOF material constructed from flexible ligands for use as the cathode material in aqueous zinc-ion batteries. By modifying the material morphology and increasing the interlayer spacing, the zinc-ion storage capacity was significantly enhanced, achieving outstanding rate capability and long-term cycling stability. The material delivered a high specific capacity of 369.8 mAh g⁻¹ at a current density of 500 mA g⁻¹ and maintained stability over 640 cycles. Comprehensive characterization and detailed density functional theory (DFT) calculations further elucidated its charge storage mechanism.



Scheme 1. Schematic of the preparation process for Zn-BBA@PANI cathode.

2. Results and discussion

2.1. Synthesis and structural characterization

Zn-BBA was synthesized via the hydrothermal method reported in the literature.[36] In this study, PANI was successfully intercalated into the interlayer space of the 2D layered material Zn-BBA, which is composed of flexible ligands, resulting in the synthesis of the Zn-BBA@PANI composite material (**Scheme 1**). Zn-BBA crystallizes in the monoclinic $C2/m$ space group, with its crystal structure featuring binuclear zinc clusters as fundamental building blocks. These clusters coordinate with flexible ligands to form a regular 2D layered framework (**Figure 1a**). Adjacent 2D layers are further extended into a 3D structure via weak π - π stacking interactions, with an interlayer distance of 0.48 nm (**Figure 1b** and **Figure 1c**). This loosely coupled interlayer structure provides an ideal spatial positioning and structural foundation for PANI intercalation.

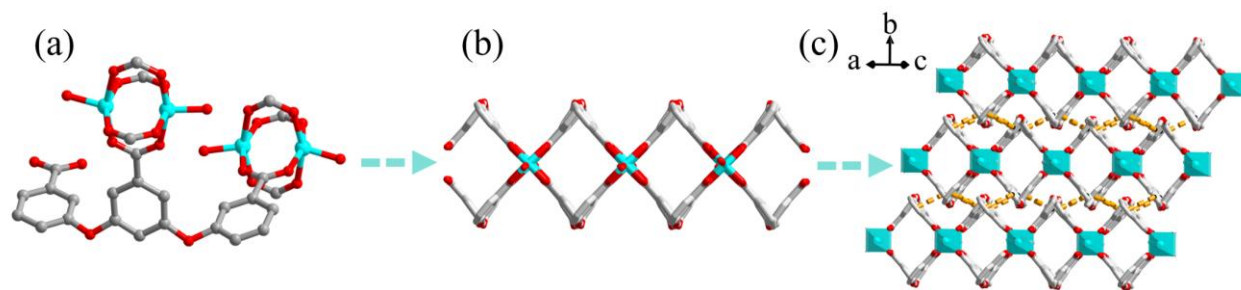


Figure 1. Structural Features of Zn-BBA. (a) Coordination monomer diagram of Zn-BBA; (b) 2D single-layer planar schematic diagram; (c) Schematic illustrations of the 3D extension via weak interactions. (Solvent molecules are omitted for clarity. Blue spheres represent zinc (Zn), gray spheres represent carbon (C), and red spheres represent oxygen (O).)

The powder X-ray diffraction (PXRD) pattern shows that the diffraction peaks of Zn-BBA agree with the simulated crystallographic data, indicating that the synthesized Zn-BBA exhibits good phase purity (**Figure 2a**). Following PANI intercalation, the interlayer spacing of the (040) crystal plane increased from 0.48 nm to 0.68 nm (**Figure S1**). Additionally, the intensity of diffraction peaks noticeably diminished, indicating partial amorphization of the material. Most diffraction peaks remained consistent with Zn-BBA, while new peaks appeared at 15.2° and 44.2°, corresponding to characteristic peaks of PANI. This confirms the successful intercalation of PANI. The successful intercalation of PANI in Zn-BBA was further confirmed using Thermogravimetric analysis (TGA) and Fourier transform infrared spectroscopy (FT-IR). The TGA curve clearly reveals that Zn-BBA@PANI exhibits two weight losses below 350 °C, corresponding to weight loss rates of 3% and 5%, respectively. This is attributed to the evaporation of physically adsorbed water and DMF molecules within the structure (**Figure 2b**). The weight loss of 22% below 400 °C corresponds to the evaporation and thermal decomposition of PANI, while the 12% weight loss below 550 °C reflects the gradual collapse of the structural framework. Zn-BBA exhibits gradual structural collapse at 400 °C, corresponding to a weight loss of 29%. These results indicate that the thermal stability of the material is enhanced following successful PANI intercalation. The FT-IR test results indicate, the C=O stretching vibration peak of H₃BBA completely disappears at 1686 cm⁻¹ (**Figure 2c**), this confirms the successful coordination of Zn²⁺ with H₃BBA. In the Zn-BBA@PANI composite, newly observed peaks at 1457 cm⁻¹ and 1091 cm⁻¹ correspond to the C-N and C-H stretching vibrations of the PANI, respectively, confirming the successful incorporation of PANI into the Zn-BBA layers.

The microstructure of Zn-BBA@PANI is shown in **Figure 2d** and **Figure S2**, revealing a block-like structure. Compared to Zn-BBA (**Figure 2e**), the morphology of Zn-BBA@PANI undergoes

a transformation. Upon PANI intercalation, a nanoporous structure characterized by nanoparticle agglomeration, reduced particle size, and increased surface roughness is formed. This structural transformation is primarily attributed to the oxidative polymerization reaction of aniline molecules in the interlayer regions of the Zn-BBA@PANI composite. This structural change promotes the exposure of active sites and enhances mass transport processes. Energy-dispersive X-ray spectroscopy (EDS) images and elemental distribution maps confirm the uniform distribution of Zn, O, and N elements within Zn-BBA@PANI (**Figure 2f**).

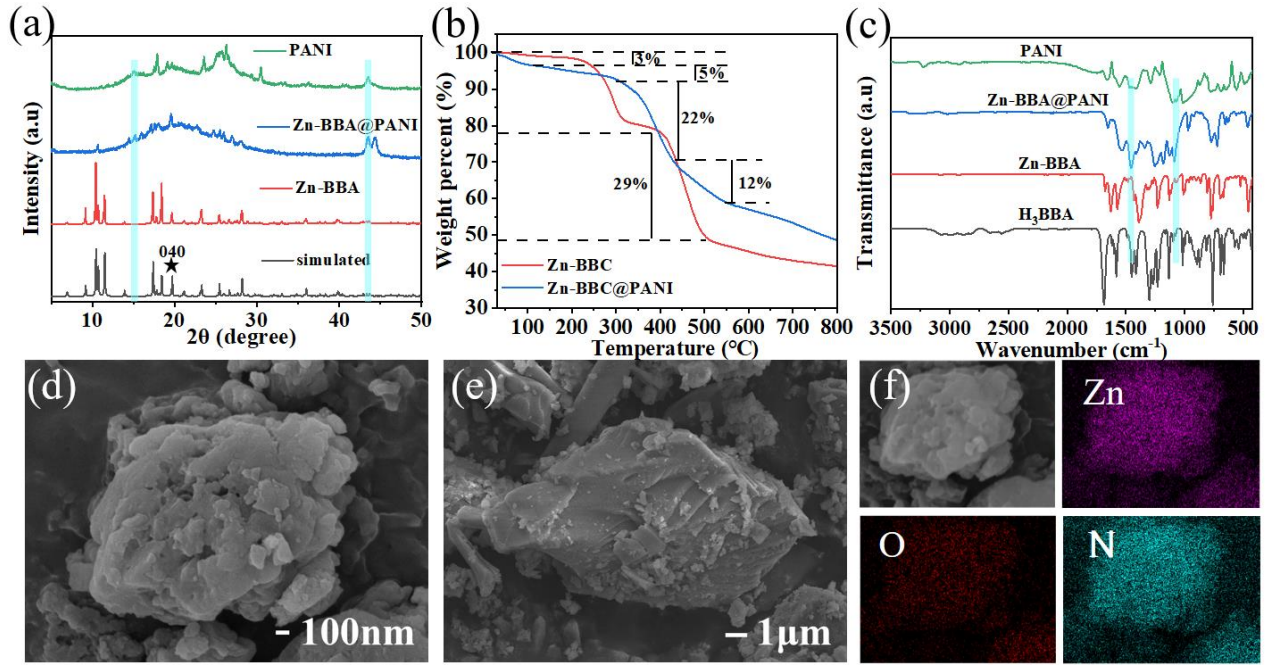


Figure 2. Structural characterizations and morphological of the Zn-BBA and Zn-BBA@PANI: (a) PXRD patterns of Zn-BBA and Zn-BBA@PANI; (b) TGA curves of Zn-BBA and Zn-BBA@PANI; (c) FT-IR spectra of Zn-BBA and H₃BBA, Zn-BBA, Zn-BBA@PANI, and PANI; (d) SEM image of Zn-BBA; (e) SEM image of the Zn-BBA@PANI; (f) EDS of the Zn-BBA@PANI

2.2. Electrochemical characterization

AZIBs were assembled using Zn-BBA@PANI as cathode materials and tested for their electrochemical performance. The redox behavior of Zn-BBA@PANI was evaluated using cyclic voltammetry (CV) curves. At a scan rate of 10 mV s^{-1} , the first three cycles demonstrate excellent overlap in the cyclic voltammetric curves of Zn-BBA@PANI, indicating highly reversible reaction characteristics and stable cycling performance in AZIBs (**Figure 3a**). Zn-BBA also exhibits a relatively overlapping trend (**Figure S3**). In addition, at a current density of 0.5 A g^{-1} , the first five cycles of the Zn-BBA@PANI electrostatic discharge-charge curve exhibits distinct charge-discharge voltage plateaus corresponding to the redox potentials of the CV curve, which are associated with the insertion/extraction of active Zn^{2+} (**Figure 3b**).

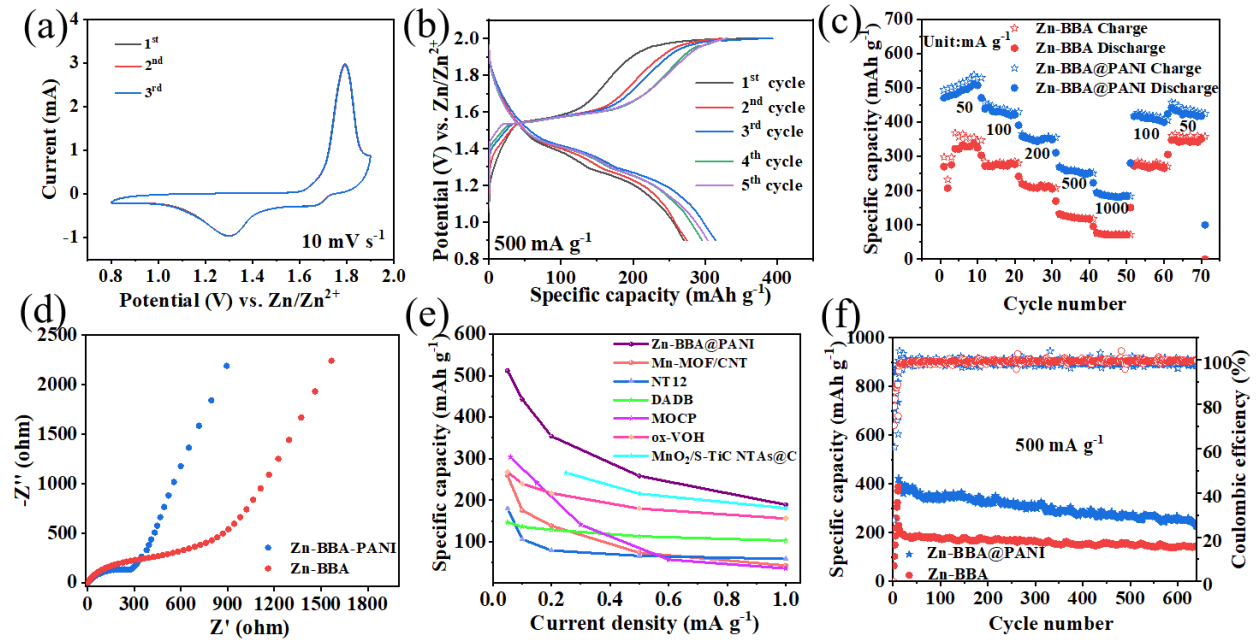


Figure 3. (a) CV curves of Zn-BBA@PANI at 10 mV s^{-1} ; (b) GCD curves at 500 mA g^{-1} during the first five cycles of Zn-BBA@PANI; (c) Multiplicity performance of Zn-BBA and Zn-BBA@PANI at different current densities; (d) EIS plots of Zn-BBA and Zn-BBA@PANI; (e) Comparison of rate capacities between the present Zn-BBA@PANI and the previously reported MOF-based cathodes for AZIBs; (f) Cycling performance and Coulombic efficiency of Zn-TCT and Zn-TCT@CNT at a current density of 500 mA g^{-1}

The Zn-BBA@PANI electrode exhibits discharge specific capacities of 511.7, 443.4, 354.1, 258.4, and 189.3 mAh g⁻¹ at current densities of 50, 100, 200, 500, and 1000 mA g⁻¹ sequentially. Even at a current density of 1 A g⁻¹, the reversible discharge specific capacity of the Zn-BBA@PANI electrode can still reach 189.3 mAh g⁻¹. When the current density was restored to 50 mA g⁻¹, the capacity of the Zn-BBA@PANI electrode recovered to 431.4 mAh g⁻¹, demonstrating excellent rate performance (**Figure 3c**). In contrast, the capacity of the Zn-BBA electrode at 50, 100, 200, 500, and 1000 mA g⁻¹ was 337.7, 270.9, 213.9, 123.2, and 73.7 mAh g⁻¹, sequentially. The reaction kinetics of the obtained electrode were further studied by EIS measurements. The semicircle in the high-frequency region for Zn-BBA@PANI is significantly smaller than Zn-BBA, indicating that the charge transfer resistance (R_{ct}) of Zn-BBA@PANI is lower. In the low-frequency region, the slope of the straight line for Zn-BBA@PANI is significantly steeper than that for Zn-BBA, corresponding to the ion diffusion coefficient, further indicating that the Zn-BBA@PANI cathode exhibits faster Zn²⁺ storage kinetics (**Figure 3d**). In addition, the electrochemical performance of Zn-BBA@PANI is superior to that of most previously reported MOF-based cathodes, indicating that Zn-BBA@PANI is a promising AZIBs cathode (**Figure 3e** and **Table S1**).^[37–42] The long-term stability of the Zn-BBA@PANI electrode was evaluated at a current density of 500 mA g⁻¹. After 640 cycles, the specific capacities of Zn-BBA@PANI and Zn-BBA were 219.4 mAh g⁻¹ and 99.78 mAh g⁻¹, correspondingly (**Figure 3f**).

To investigate the Zn²⁺ storage characteristics of the Zn-BBA@PANI and Zn-BBA cathode materials, CV was employed to test the Zn-BBA@PANI electrode at different scan rates (**Figure 4a**).^[43] A pair of redox peaks appears at 1.7 V and 1.3 V, corresponding to the insertion/extraction of Zn²⁺ during the discharge phase. All curve profiles exhibit similar shapes to the redox peaks and gradually increase with rising scan rates, indicating the battery possesses fast electrochemical

response and excellent reversibility. Zn-BBA exhibits distinct redox peaks at 1.64 V and 1.32 V (Figure S4). The b-values (Formulas S1 and S2) are used to quantify the degree of pseudocapacitive contribution,[44] the b-values for the cathodic peak (b_1) and anodic peak (b_2) were calculated as 0.87 and 0.65, respectively (Figure 4b and Figure S5). These results indicate that the Zn^{2+} storage process in the Zn-BBA@PANI electrode is primarily controlled by a mixed mechanism of capacitive response and diffusion process. Furthermore, the capacitive contributions and diffusion-controlled contributions of Zn-BBA@PANI and Zn-BBA cathodes at different scan rates were calculated using Formula S3 (Figures 4d and S7). As the scan rate increased from 2 mV s^{-1} to 10 mV s^{-1} , the capacitive contribution correspondingly rose from 65% to 80% (Figure 4d and Figure S7), indicating that the electrochemical behavior of the Zn-BBA@PANI electrode is primarily dominated by capacitive effects. At a specific scan rate (10 mV s^{-1}), cyclic voltammetric curves were fitted (Figure 4c and Figure S6), where the shaded regions represent capacitive contributions, reaching up to 80%, this capacitance-dominated energy storage mechanism facilitates rapid charge/discharge kinetics. The diffusion coefficient of Zn^{2+} in Zn-BBA and Zn-BBA@PANI cathodes was determined using the GITT method (Formula S4). After embedding Zn-BBA into the PANI cathode, the specific capacity of the electrode significantly increased, indicating enhanced storage performance for Zn^{2+} (Figure 4e). Concurrently, the Zn^{2+} diffusion coefficient of the Zn-BBA@PANI cathode was higher than that of the pure-phase Zn-BBA cathode (Figure 4f and Figure S8). This outstanding electrochemical performance stems from the interlayer spacing expansion induced by PANI intercalation and its unique π -conjugated structure. The synergistic effect of these two factors effectively weakens the electrostatic interactions between Zn^{2+} and the host framework, thereby significantly enhancing the migration dynamics of Zn^{2+} within the layered structure.

In addition, the reason for the excellent Zn^{2+} storage performance of the Zn-BBA@PANI electrode was further analyzed by Density-functional theory (DFT) calculations, and the bandgap width was significantly reduced after the introduction of PANI as compared to pure Zn-BBA (**Figure 4g** and **Figure S9**), indicating that the improved intrinsic electronic conductivity facilitates the enhancement of the electron transfer ability and exhibits excellent multiplicity performance. Subsequently, the molecular electrostatic potential (MESP) distribution of Zn-BBA was calculated using the Zn-BBA molecule as a model to identify potential zinc storage active sites (**Figure 4h**). As shown, red regions indicate positive MESP values (i.e., nucleophilic centers), while blue regions denote negative MESP values (i.e., electrophilic centers). Consequently, the oxygen atom of the carboxyl group in Zn-BBA, exhibiting negative MESP, demonstrates high reactivity toward absorbing Zn^{2+} . To elucidate the charge transfer of Zn^{2+} , the differential charge density of Zn^{2+} inserted into Zn-BBA@PANI was also analyzed (**Figure 4i**). Yellow and blue regions indicate electron density enrichment and depletion, correspondingly, primarily localized within the benzene rings of the PANI molecules. This demonstrates that the tunable interlayer spacing induced by PANI intercalation, coupled with its unique π -conjugated structure, effectively suppresses electrostatic interactions between Zn^{2+} and the host framework. Consequently, this significantly enhances the migration dynamics of Zn^{2+} within the layered structure

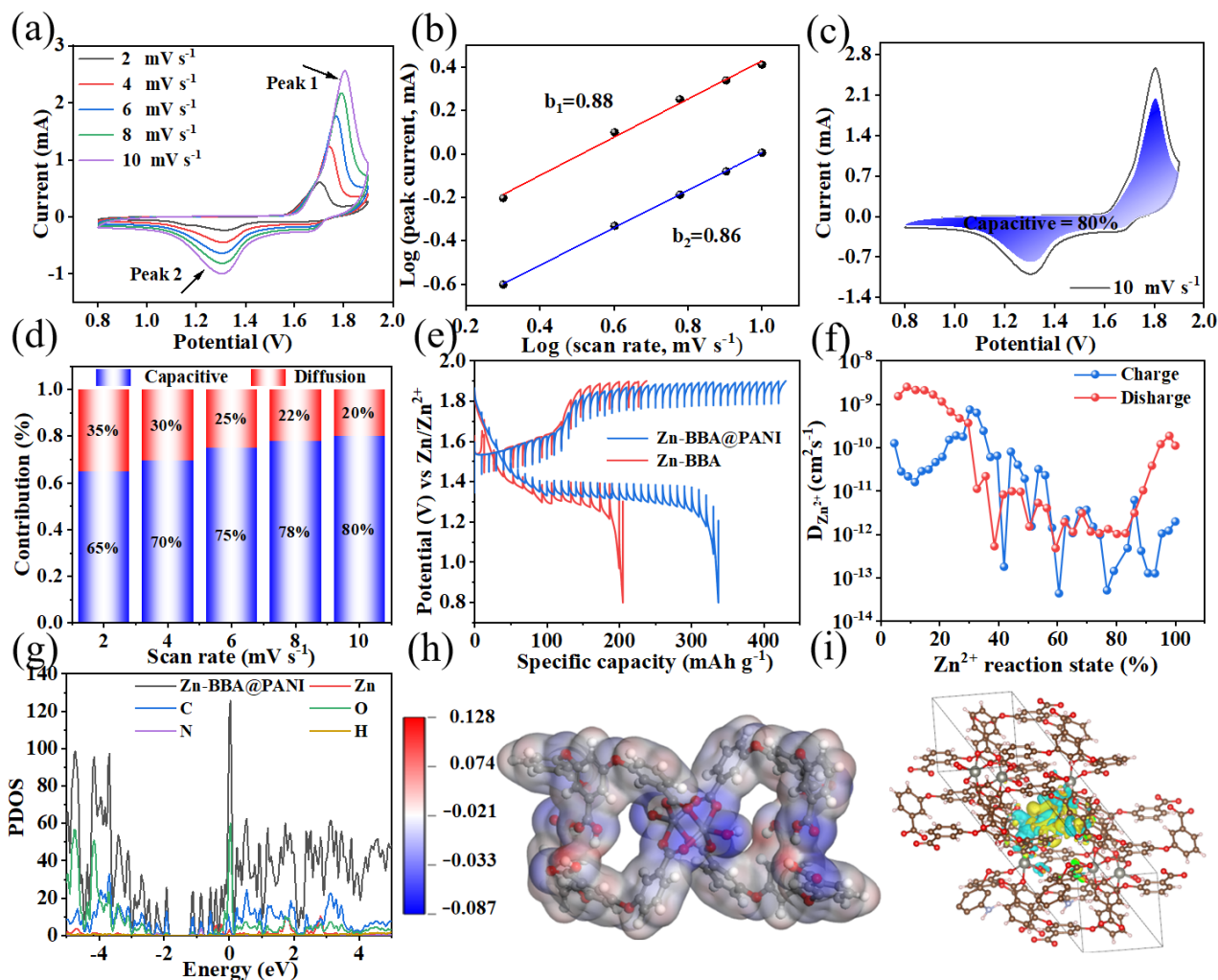


Figure 4. (a) CV curves of Zn-BBA@PANI cathode at different scan rates from 2 to 10 mV s⁻¹; (b) The corresponding log (peak current) versus log (scan rate) plots at each redox peak for Zn-BBA@PANI cathode; (c) Contributions of capacitance-controlled capacity (the blue part) and diffusion-controlled capacity (the interspace part) of Zn-BBA@PANI at 10 mV s⁻¹; (d) Bar chart of the contributions between capacitance-controlled capacity and diffusion-controlled capacity of Zn-BBA@PANI at different scanning rates; (e) Comparison diagram of the charge-discharge GITT curves between Zn-BBA@PANI and Zn-BBA; (f) Calculated Zn²⁺ diffusion coefficient ($D_{Zn^{2+}}$) versus Zn-BBA@PANI capacity curve; (g) PDOS of Zn-BBA@PANI; (h) Electrostatic potential mapping of Zn-BBA; (i) Differential charge density of Zn-BBA@PANI

2.3. Mechanisms for electrochemical energy storage

To elucidate the storage mechanism of the Zn-BBA@PANI cathode, PXRD, X-ray photoelectron spectroscopy (XPS), and Transmission electron microscopy (TEM) analyses were performed to investigate its structural and morphological evolution during cycling. To analyze the variation of the Zn-BBA@PANI interlayer spacing during charge-discharge cycling, PXRD measurements were performed on the electrode in different charge-discharge states after three cycles (**Figure 5b** and **Figure 5c**). **Figure 5a** shows a magnified view revealing a prominent characteristic peak at 16.5° . Notably, reversible shifts in the diffraction peak were observed during battery cycling, indicating that the reversible change in interlayer spacing results from the insertion/extraction of Zn^{2+} . During discharge, new diffraction peaks at 8.5° and 28.7° appeared in the material's PXRD spectrum, corresponding to the formation of a heterogeneous phase of $\text{Zn}_4\text{SO}_4(\text{OH})_6 \cdot 4\text{H}_2\text{O}$ (corresponding to PDF#44-0673). These peaks reached maximum intensity at 0.8 V, confirming phase formation, upon voltage recovery to 1.8 V, the new phase gradually disappeared, indicating that its formation and dissolution constitute a reversible process.

XPS was employed to investigate the valence state changes of Zn-BBA@PANI during cycling. The presence of Zn, O, and C elements is detected in both charged and discharged states (**Figure 5f**). When the Zn-BBA@PANI/Zn battery is discharged to 0.8 V, two characteristic peaks corresponding to $\text{Zn}2\text{p}_{3/2}$ and $\text{Zn}2\text{p}_{1/2}$ are observed, indicating the successful insertion of Zn^{2+} into the Zn-BBA@PANI framework. Upon charging the electrode to 1.9 V, the intensity of the $\text{Zn}2\text{p}$ peaks decreases. The $\text{Zn}2\text{p}$ signal detected during charging may originate from residual adsorbed Zn^{2+} on the surface of the Zn-BBA@PANI electrode. This observation confirms that Zn^{2+} can reversibly insert into and extract from the Zn-BBA@PANI structure during charge-discharge cycles (**Figure 5e**). A similar trend is also observed in the O 1s spectrum **Figure 5d**). Compared

with the initial and fully charged states, the areas of the C=O and O–C peaks are larger in the fully discharged state. SEM was further used to monitor the morphological development of the Zn-BBA@PANI electrode. Compared with the initial state (**Figure 5g**), the cathode surface is covered with irregular $\text{Zn}_4\text{SO}_4(\text{OH})_6 \cdot 4\text{H}_2\text{O}$ nanosheets in the fully discharged state (**Figure 5i**). These nanosheets gradually disappear during the subsequent charging process (**Figure 5h**). The highly reversible morphological transformation during charge-discharge cycles is in good agreement with the phase progression observed in the PXRD patterns, further confirming the reversibility of the electrode reaction.

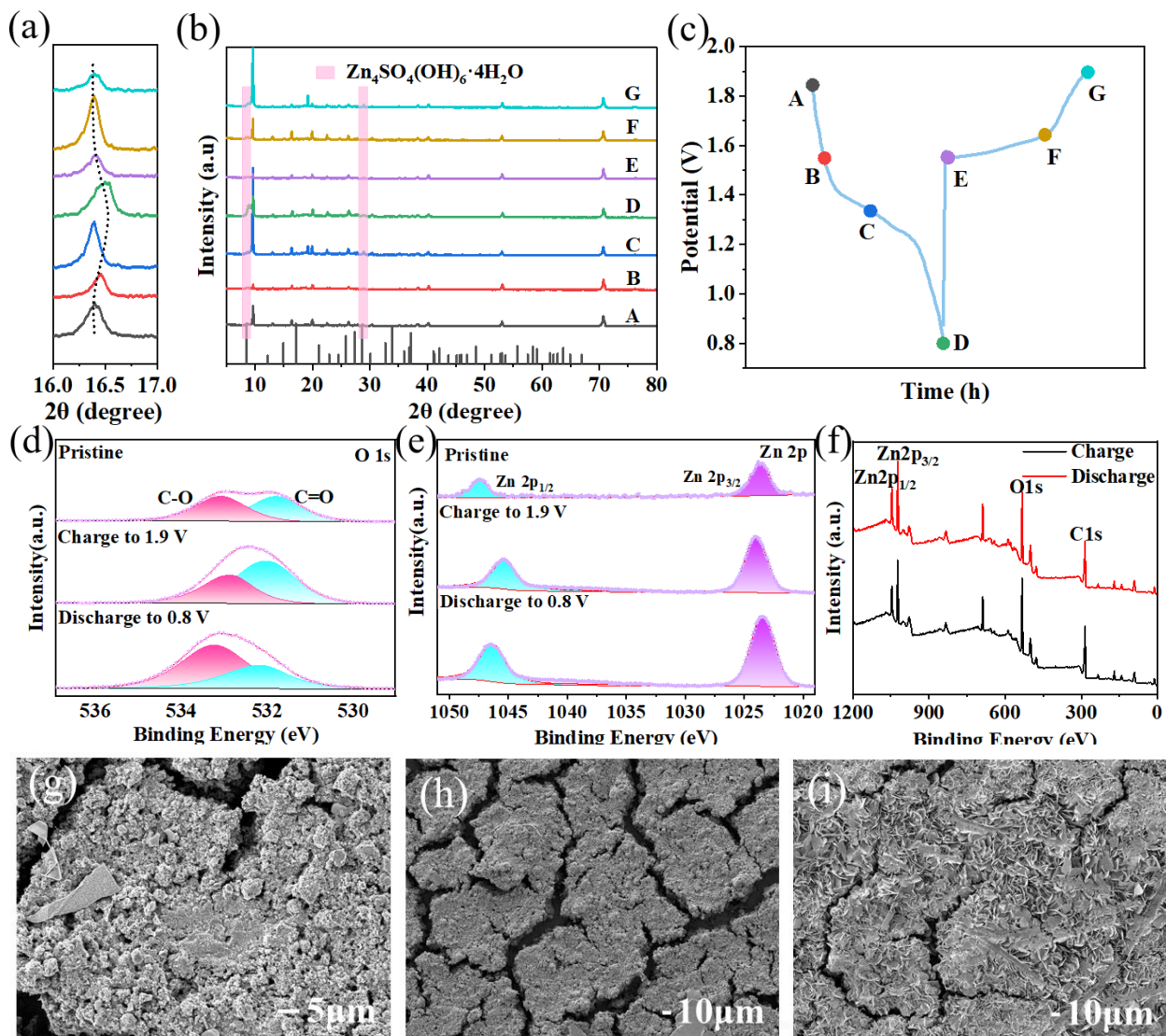


Figure 5. Charge storage behavior of Zn-BBA@PANI after the third cycle: (a, b, c) Selected points for comparative analysis and evolution of XRD patterns collected under different states, along with GCD curves; (d) XPS spectrum of O1s; (e) XPS spectrum of Zn2p; (f) Full XPS spectrum; SEM images of the Zn-BLB cathode: (g) Initial state, (h) Fully charged state, (i) Fully discharged state.

3. Conclusions

In summary, by intercalating PANI molecules with a small molecular size and a unique π -conjugated structure into Zn-BBA with a large interlayer spacing constructed from flexible ligand, a composite material with expanded interlayer spacing was successfully designed. The distinctive structure of PANI and the intercalation-induced enlargement of the interlayer spacing, synergistically enhances the electrical conductivity of the material and furnishes favorable channels for Zn^{2+} diffusion. Furthermore, the partial amorphization of the material after PANI intercalation creates abundant active sites, further optimizing its electrochemical kinetics. This leads to a significant performance enhancement, with Zn-BBA@PANI exhibiting a high specific capacity of 219.4 mAh g^{-1} following 640 cycles at 500 mA g^{-1} and achieving 189.3 mAh g^{-1} at 1 A g^{-1} , thereby demonstrating outstanding rate capability. This study proposes an effective strategy for embedding PANI into flexible 2D layered MOF materials. By modulating the interlayer spacing and incorporating conductive polymers, it successfully addresses the critical challenges of ion transport and conductivity, thereby opening a new avenue for developing high-performance 2D MOF cathodes in aqueous zinc-ion batteries.

Author Contributions

Siyu Shi: Conceptualization, Methodology, Writing – original draft, Writing – review & editing.
Shuqi Xu: Data curation, Methodology. Mingchang Zhu: Conceptualization, Funding acquisition.
Ying Zhang: Investigation. Yaguang Sun: Conceptualization. Lei Wang: Investigation. Jingwei
Liu: Supervision, Methodology. Shuangyan Wu: Conceptualization, Supervision, Funding
acquisition. Junqi Su: Formal analysis, Conceptualization.

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supporting plan for “Outstanding youth” of SYUCT (2022YQ010), The Basic Research Project of
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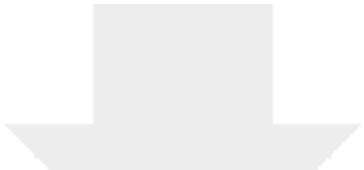
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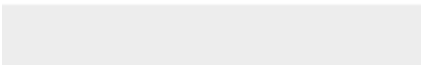
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Highlights

- (1) PANI-intercalated 2D flexible MOFs as cathode for aqueous zinc-ion batteries.
- (2) Intercalation boosts amorphization, providing more active sites and better capacity.
- (3) Zn-BBA@PANI delivers 219.4 mAh g^{-1} at 0.5 A g^{-1} over 640 cycles.
- (4) PANI narrows DOS band gap and boosts active sites.

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: